# Pavan Kumar

**LeetCode** | **Codeforces** | **CodeChef** | **Kaggle** | **AtCoder**
Computer Science and Engineering
AIML

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## Skills

- **Languages** : C++, C, Python 3, Java 8+, Kotlin, MongoDB, JavaScript, SQL
- **Frameworks & Libraries:** Flask, Gradio, scikit-learn, Lang Chain, NumPy, Pandas, Tensorflow, spaCy, NLTK
- **Developer Tools** : Git, GitHub, Docker, SupaBase, Postman, Jupyter Notebook, Android Studio, Cisco PKT
- **CS Fundamentals** : OOPs, Operating Systems, Computer Networks, DBMS, Statistics, Agile Methodology
- **Soft Skills** : Leadership, Collaboration, Adaptability, Communication, Productivity, Discipline, Consistency

## Experience

- **AI Performance Engineer | Action Extension v2 | Deccan AI Experts** March `26
  - Fine-tuned **large language models (LLMs)** using supervised and instruction tuning, achieving **+18% improvement in ROUGE-L** and **+15% increase in BLEU score** on summarization and generation tasks.
  - Evaluated model outputs using **ROUGE (ROUGE-1, ROUGE-2, ROUGE-L)** and **BLEU** metrics, consistently improving text quality and alignment with reference outputs.
  - Conducted **error analysis** on low-scoring samples (ROUGE/BLEU), reducing hallucinated or irrelevant outputs by **~22%** through targeted fine-tuning.
  - **Key Skills:** LLM Fine-Tuning, ROUGE, BLEU, Perplexity, NLP Evaluation, Transformers, PyTorch, Generative AI.

## Projects

- **Credit Risk Assessment Using Machine Learning | GitHub** Nov `25
  - Architected a machine learning pipeline to analyze **10k+ loan applications**, predicting default likelihood with high precision.
  - Executed data processing on **1:10 imbalanced data**, using **SMOTE** to boost minority class representation by 40%.
  - Optimized model selection across 4+ complex algorithms, with **XGBoost** significantly outperforming the baseline **Logistic Regression** by **12% in F1-score** across all test sets.
  - Surpassed benchmarks with a **92.8% AUC**, reducing missed defaults by 15% compared to standard models.
  - **Tech:** Python, Scikit-learn, Pandas, Matplotlib, Seaborn, Numpy, Pickle.
- **UDP-Based DNS Query Handling | GitHub** July `25
  - Engineered a functional DNS resolver from scratch, **handling 100+ concurrent UDP queries** using core protocol logic.
  - Constructed a Trie-based lookup system, reducing resolution time to **under 5ms** for high-speed record retrieval.
  - Integrated an LRU caching layer with a 65% hit rate, cutting repeated upstream lookups by more than half.
  - **Tech:** Retrieval Trees, HashMap, Strings, LRU Cache.
- **Apache Kafka Broker | GitHub** June `25
  - Spearheaded a C++ message broker, achieving a sustained throughput of **5,000+ messages** per second.
  - Developed **log persistence**, **offset management**, and **multi-topic message retrieval** for high reliability.
  - Mastered low-level systems programming concepts including **TCP socket communication** and binary serialization in Java to optimize packet delivery and maximize overall network efficiency.
  - Tech: Java, TCP sockets, Kafka protocol, Networking, Serialization, Log Storage.
- **Handwritten Digit Recognition using Deep Learning | GitHub** Apr `25
  - Pioneered a CNN classification system, processing the **60,000-image MNIST dataset** to identify handwritten digits with human-level accuracy and minimal computational overhead.
  - Implemented the model in Python using **TensorFlow**, performing **data preprocessing and tuning**.
  - Attained **99% accuracy** and deployed a real-time digit prediction UI for seamless interaction.
  - **Tech:** Python, TensorFlow/Keras, NumPy, Matplotlib, Seaborn, Jupyter Notebook.

## Certificate

- MongoDB Fundamentals | Completed hands-on NoSQL training | GitHub Learning Nov `25
- Design & Analysis of Algorithms | Accomplished top 5% performance | NPTEL Feb `25

## Achievements

- LeetCode | Max Rating 1609 | Maintaining a 485+ days continuous solving streak | Solved 800+ Problems
- CodeChef | Max Rating 1605 | 3 STAR Coder | Division 2 | Global Ranks: 235, 381, 637, 656, 820 | Attended 20+ Contests
- GeeksForGeeks | 280+ Coding Score | Maintaining a 50+ days POTD streak | Solved 60+ Problems
- GitHub | 3,185+ Contributions (Last 12 Months) | Consistent Commit Streak
- Coding Ninjas | Max Rating 1808 | Specialist | Grand Master | Attended 10+ Contests

## Education

| Degree/Board | Institution | Year | CGPA/% |
|---|---|---|---|
| B.Tech, CSE-AIML | Lovely Professional University | 2023 - Present | 9.38 |
| BIEAP - XII | Narayana Junior College | 2021 - 2023 | 98.1% |
| BSEAP – X | Narayana Group of Schools | 2020 - 2021 | 99% |